

Re-Shaping the EU Digital Identity Framework

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ABSTRACT

Electronic authentication and digital signature are the base components of the European Union (EU) Digital Single Market. The area is regulated by the eIDAS (electronic identification and trust services for electronic transactions in the internal market) regulation that is compulsory for all Member States since 2018. Despite the Member States' efforts, the regulation implementation has not been as successful as expected. Therefore, the European Commission initiated the eIDAS revision process in the second half of 2020. Based on the collected feedback, the Commission proposed in July 2021 the first draft of the renewed eIDAS regulation establishing the European Digital Identity framework. The aim of this research is to analyze the feedback provided by different countries and sectors in the eIDAS review process (156 pages of material) and evaluate their correspondence to the Commission proposal. The research follows the exploratory case study methodology and we use thematic analysis for the evaluation. The outcome of this study shows whether all relevant expectations of the interested parties are covered by the Digital Identity Framework proposal and the research results are a valuable input for the Commission as the debate over the eIDAS regulation draft is ongoing.

CCS CONCEPTS

• **Social and professional topics** → **Government technology policy**.

KEYWORDS

eIDAS, electronic authentication, electronic identity, implementation challenges, identity management, digital signature

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1 INTRODUCTION

The European Digital Single Market is one of the backbones of the EU and part of the EU single market strategy ¹. The Digital Single Market largely depends on the electronic identities of Member States and provides trust services regulated by the eIDAS, electronic identification and trust services for electronic transactions in the internal market, regulation. EU citizens need to be able to use their national eIDs for the services in other EU Member States. Meanwhile, there should be a guarantee that different electronic authentication schemes operate across borders without limitations inside the EU and given digital signatures are legally binding in all Member States [21]. According to the European Commission eIDAS impact assessment report, the ongoing COVID pandemic situation has sped up the need for digitization by seven years ². This means that the need for digitization and the importance of a well functioning EU Digital Single Market has become even more relevant.

To enable an interoperable and secure e-service provision, cross-border authentication, and acceptance of electronic signatures, the EC adopted, on the 23rd of July 2014, the regulation on electronic identification and trust services for electronic transactions in the internal market and repealing Directive 1999/93/EC (eIDAS) ³. Until 2016, the adoption of this regulation was voluntary for the Member States. Since September 2018 it has become mandatory [12].

Unfortunately, despite the Member States' efforts, the implementation of the eIDAS regulation has not been simple, and the EU digital market is not operating as expected [12]. Every Member State has its individual digital market operating within the national legal framework. This variety of different digital identification solutions in the EU has gradually become an obstacle in terms of cross-border interoperability in the European internal market [17, 22]. In four years since mandatory implementation, only 59% of the EU population has a chance to benefit from the EU Digital

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¹Single Market and Standard. https://ec.europa.eu/growth/single-market_en

²European Commission. eIDAS Impact Assessment Report. <https://ec.europa.eu/newsroom/dae/redirection/document/76618>

³eIDAS regulation. Available: <http://data.europa.eu/eli/reg/2014/910/oj>

Single Market⁴. Therefore, the EC initiated, in the second half of 2020, the eIDAS public consultation process to collect the interested parties' feedback about the existing regulation and related challenges. The aim was to revise the regulation. After analysing the feedback, the EC proposed, in July 2021, an amended version of the eIDAS regulation, which included the European digital identity framework.

This research focuses on the analysis of the EC's proposal and its correspondence to the feedback received during the public consultation procedure. The aim is to find out, whether the proposal covers public and private sector authorities' main concerns towards the eIDAS regulation, and to provide input to the ongoing eIDAS revision process.

Based on the aim of this research, the main research questions are:

- What are the public and private sector authorities' main amendment expectations towards the eIDAS regulation?
- How do the eIDAS regulation amendments proposed by the European Commission meet the public and private sector authorities' expectations?

Sect. 2 provides an overview of electronic identity and eIDAS related literature. Sect. 3 gives an overview of the eIDAS regulation and its state of play, including the EC eIDAS revision proposal. Sect. 4 describes the research methodology and data collection procedure and analysis method. Sect. 5 presents the research findings from the private and public sector perspective. In Sect. 6, we analyze the correlation of interested parties' feedback to the eIDAS regulation proposal. Sect. 7 provides an insight to the research limitations and to the future research perspective and we conclude the paper in Sect. 8.

2 LITERATURE REVIEW

The electronic identity is a central concept for the development and operation of digital government [8, 23] and e-commerce [15]. For instance, the eID became a part of the Estonian critical infrastructure [22], and the state itself is the eID primary end-user and highly dependent on its eID [25]. This section provides an overview of the eID and eIDAS related literature that is relevant from this research perspective with the focus on the obstacles and triggers of the EU digital identity implementation from the public and private sector authorities' perspective.

Despite the importance of the concept, there is no universal definition for electronic identity. The literature indicates broad and narrow concepts of identity in the digital space. For instance, Hoikkaen defines the term eidentity as a "data set related to a personal or collective identity stored and transferred in the electronic systems" [7]. It is worth mentioning that Hoikkaen uses the terms electronic identity, digital identity and eID interchangeably [7]. At the same time, Khatchatourov uses the term eID only in the context of the eIDAS Regulation [8]. Contrary, van Dijck and Jacobs specify eIDs as "digital solutions to prove one's identity", where the main functionalities of the solutions comprise authentication, login and digital signing [26]. Overall, identity in the digital space can relate to all online transactions [8]. Current work focuses on the national

electronic identification (eID) and authentication schemes and trust services as part of the EU Digital Single Market.

The topic of eIDAS implementation processes in EU countries is relatively new and only partially researched. Researchers focus mainly on some specific areas or technical issues of the eIDAS [12]. For instance, among technical solutions that are examined in the framework of eID systems and eIDAS: the authentication of additional data and different cryptography solutions [14], pseudonyms and pseudonymous signature [8, 11], and integration of block-chain technology with Qualified Electronic Signatures [24]. Further, there are proposals to widen the scope of the technological solution. For example, including an electronic signature of the ICO smart contracts [27].

Furthermore, the literature analyses the national eID systems, their integration with eIDAS-Node, their further extensions, proposes alternative technical solutions. For instance, the German eID schema is broadly examined [11, 13, 14]. Further, an overview of the Italian architecture for the eIDAS-Node and connection of the eID scheme is provided [21], and the Dutch IRMA eID system is outlined [26]. Some research papers offer various applications of the eIDAS-Node in an educational context [3, 5, 6, 9]. For example, proposing an extension of the basic set of attributes by adding academic attributes as part of citizens' profiles and offer technical solutions. They claim that it would be beneficial for citizens in the education context. In particular, this would save the students' time on the application procedure and allow them to use academic services through national eID [3].

Since the provision of online services is closely related to the concerns of "security, privacy, and trust" [2] moreover, a multitude of various digital identities brings inconveniences for users and endangers their security and privacy in cyberspace [15], privacy aspects are widely discussed in the context of electronic identity, identity management and eIDAS [8, 11, 14]. For example, Kim Nguyen considers aspects of trust that are embedded in the eIDAS regulation. Firstly, he argues that the certification procedure guarantees users that the provider's services are trustworthy. The requirements for the trust services provision, the systems itself and its' elements are established in the European standards. Moreover, the certification is provided by independent third parties and supervised by national agencies. As a result of certification, all qualified trust service providers are registered in the trust lists together with the description of their services. Other criteria that would ensure trust by Nguyen are the "evaluation of the cryptography process, the definition of minimal requirements, and decentralized trust models based on transparency principles" [16].

Some researchers investigated if eIDAS is beneficial for the Member States and their national cross-border programs and e-government objectives or somewhat burdensome. Although eIDAS poses additional obligations on the Member States, they suggest that it rather supports national initiatives and projects, such as e-residency in Estonia, than challenges them. Therefore, it is beneficial for the governments to implement eIDAS [1].

Nevertheless, initial research on eIDAS implementation indicates that some countries are more successful in their endeavours, while others are hesitant and struggle with the eIDAS implementation [17]. Early comparison of national eID systems in Europe demonstrates that there are different technical and organisational

⁴European Commission. eIDAS Impact Assessment Report. <https://ec.europa.eu/newsroom/dae/redirection/document/76618>

elements between the systems [10], also architectural solutions of the identity systems are diverse [8]. The reason for this diversity can be clarified by the fact that each EU Member State developed their eID management system independently [21], based on the earlier systems and during "incremental innovation" and "path continuation" [10]. Each country tried to meet its' internal goals to provide secure authentication, while "interoperability with other state's eID schemes was no priority" [19].

However, identification and authentication systems of different EU countries have many similarities [20]. The diversity of the rules and systems in the electronic identity management between countries caused issues with interoperability and turned out to be an obstacle for cross-border electronic services and operation of the EU Digital Single Market [19, 21].

Generally, information systems operate on identities that connect citizens with the digital information stored in the databases. If identifiers in different databases vary, cross-referencing the information from one database to another is hindered. Therefore, the main challenge for identity management is to adapt the systems, making them interoperable and enabling cross-referencing and matching the information [4]. In other words, Member States need to implement national gateways, called eIDAS-Node, to connect to the eID systems of the other Member States [21].

Besides interoperability issues, some authors suggest that the difficulties with eIDAS implementation might be caused by the complexity of the eID concept, which encompasses more than outlined by the EU frameworks. Meanwhile, the legislation concentrates mainly on technical and legal interoperability. Other issues of a political and social nature may cause conflicts and obstacles for the eIDAS implementation. For instance, in the case of the Swedish national eID schema, it was challenging to design a new eID system having, at the same time, already existing BankID and considering opinions of all the stakeholders involved [26].

Overall, the main challenges for the member states indicated in the literature are "compliance issues", "interpretation problems", "different practices in Member States", "cooperation and collaboration barriers", and "representation of legal person" challenges [12]. Besides, the lack of knowledge among users influences the citizens' adoption rate of national eID solutions, which negatively affects the consumption of cross border electronic services. Therefore, countries should increase awareness among citizens about national eID solutions and their benefits and provide them with necessary software and qualified certificates [20].

3 eIDAS AND STATE OF PLAY

The eIDAS regulation grounded the legal foundation for electronic transactions in the EU internal market. The aim was to build trust among consumers, businesses and public authorities in the digital environment, thus boosting electronic commerce and increasing the effectiveness of public and private digital services in the European Union⁵. The regulation was adopted in July 2014 and replaced the directive 1999/93/EC on a Community framework for electronic signatures⁶. eIDAS regulation made possible to recognise other national electronic identification schemes developed in the Member

States and uniform requirements for trust services were established [18].

eIDAS regulation entered into force step by step. Figure 1 illustrates in detail the eIDAS implementation timeline from the adoption to the latest European Digital Identity framework proposal. Starting from September 2015, the Member States could start voluntarily recognise each-others eIDs. In early 2016, the eID interoperability infrastructure was available for the Member States. From July 2016, provisions referring to trust service rules became effective. Finally, from the 29th of September 2018, eIDAS regulation was obligatory for all Member States and mutual recognition of eIDs became mandatory.

The eIDAS implementation comprises several stages that each country should follow. Firstly, a Member State should start eID pre-notification: officially inform the European Commission about its "intention to notify its eID scheme". Then a peer-review stage follows, where representatives of other Member States examine and assess the eID scheme. After the peer review stage, the country notifies the European Commission about its eID scheme. As soon as the information about notification is published in the Official Journal of the European Union (OJEU), but not later than 12 months, other Member States should recognise the notified eID scheme. Since the recognition, EU citizen can use the recognised eID across borders. Germany was the first country, who notified its eID scheme in 2017, followed next year by Estonia, Spain, Croatia, Belgium, Luxembourg and Italy. Currently, 17 Member States out of 27 passed the eID notification process and three countries (Czech Republic, Norway, Austria) peer-review process is ongoing⁷.

According to the eIDAS regulation, it was planned to revise the regulation and its implementation process by 01.07.2020⁸. The EC conducted an inception impact assessment of the eIDAS revision and published a proposal to revise the eIDAS regulation on the 23rd of July 2020. According to the Inception Impact Assessment document, the EC proposed three options:⁹

- revise and slightly update the current regulation;
- extend the effect of eIDAS to the private sector;
- launch a European Digital Identity (EUid) or combine these three solutions.

During the public consultation, the EC wanted to collect feedback about the eIDAS implementation challenges from different interested parties and wanted to clarify the direction, where to develop the eIDAS regulation. The feedback was collected 23.07.2020-03.09.2020. Based on the feedback and analysis, the EC proposed in July 2021 a revised version of eIDAS - a framework for European Digital Identity¹⁰

⁷eID User Community.

Available: <https://ec.europa.eu/cedigital/wiki/display/EIDCOMMUNITY/Overview+of+pre-notified+and+notified+eID+schemes+under+eIDAS>

⁸eIDAS regulation. Available: <http://data.europa.eu/eli/reg/2014/910/oj>

⁹Inception Impact Assessment. Available: <https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=cellar:35274ac3-cd1b-11ea-adf7-01aa75ed71a1>

¹⁰Establishing a Framework for European Digital Identity. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021PC0281>

⁵eIDAS regulation Available: <http://data.europa.eu/eli/reg/2014/910/oj>

⁶Directive 1999/93/EC. Available: <http://data.europa.eu/eli/dir/1999/93/oj>

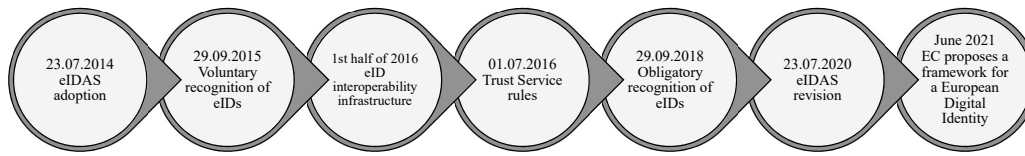


Figure 1: eIDAS timeline

3.1 European Digital Identity Framework Proposal

With the revised version of eIDAS, the EC has broadened the scope of the regulation. Previously, the regulation applied to the electronic signatures, seals, time stamps, documents and registered delivery services. In addition to that, according to the regulation draft, the eIDAS applies to the electronic archiving, attestation of attributes, remote electronic signatures and seal creation devices and electronic ledgers ¹¹.

As a major change, eIDAS proposes the legal framework for EU member states to provide EU digital identity wallets that enable users to decide, depending on the use case and needed security level, when and with whom they share various attributes (e.g. educational data, healthcare information, electronic driving license information etc.). The EU digital identity wallet should also enable electronic authentication in the online environments and giving qualified electronic signatures. The Member States must ensure the wallet solution but its usage by citizens is voluntary. The EC together with Member States will establish the technical architecture, standards and guidelines for EU digital identity wallets (also named as common toolbox) to ensure uniform approach to the wallet solution. ¹² Currently, detailed discussion over the EU digital identity wallet concept between the member states is ongoing but the schedule is intense. For example, according to the regulation draft, the Member States should implement it already in June 2024.

The eIDAS proposal specifies requirements for the remote signatures to ensure secure remote signing process. The Member States should follow the European Committee for Standardization (CEN) standards that regulate the operation and authentication to remote Qualified Signature Creation Devices. Moreover, the eIDAS draft proposal aims to harmonize with the other EU legislative initiatives like the EU directive concerning measures for a high common level of security of network and information systems across the European Union (also known as EU NIS directive) ¹³, Cybersecurity Act ¹⁴ and Single Digital Gateway regulation ¹⁵.

To summarize the key elements of the eIDAS draft proposal:

- the name eIDAS was replaced by European Digital Identity Framework with a focus on cross-border use;
- European Digital Identity Wallet concept obligatory for Member States was introduced (including Trust Mark);

- proposal enables end users sharing of different electronic attributes within the EU;
- three new qualified trust services were added to the existing list of trust services (electronic archiving, electronic ledgers and the management of remote electronic signature and seal creation devices);
- harmonisation with other EU regulations and standards.

4 RESEARCH APPROACH

The research follows exploratory case study methodology [28]. To answer the research questions, we analysed the European Digital Identity Framework Proposal and feedback provided in the public consultation process initiated by the EC and compared them. The main changes in the eIDAS regulation proposal are presented in Sect. 3.

The feedback on the Inception Impact Assessment and eIDAS regulation was collected by EC from July 2020 to October 2020 and was made available for the public on the EC website. In total, 53 responses in different formats were received from various stakeholders. Some responses contained additional downloadable documents. This research uses only the data that was publicly available.

We extracted the collected feedback from the EC website with the help of the web scraping tool Scraper and downloaded enclosed files from the web pages. In total amount 156 pages of text. Some of the feedback needed to be translated from German, Spanish, and French into the English language for further analysis. During the feedback analysis, we focused separately on two main groups of stakeholders: the public sector representatives and private sector actors of the EU Member States.

After the data extraction, we conducted a thematic analysis of the collected data sets. We conducted the analysis in four rounds using NVIVO data analysis software. Firstly, we sorted the received feedback based on the theme from which country it was sent. Figure 2 presents the detailed data analysis model.

Secondly, we split the data into three groups (case classifications: Stakeholders): the feedback from private, public organisations and others. Initially, we expected to have the most responses from private and public organisations of EU Member States. However, the third sector organisations, EU citizens, and Non-EU organisations actively participated in the public consultation process. Therefore, we formed three groups of cases: stakeholders: public sector, private sector and others.

In the third-round we tried to find a generalisation and central themes in each stakeholder group. We applied inductive data-driven approach to find patterns and probable explanations of the challenges and triggers of eIDAS implementation and stakeholders possible expectations. During the final round, we analysed every

¹¹Establishing a Framework for European Digital Identity. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021PC0281>

¹²Establishing a Framework for European Digital Identity. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021PC0281>

¹³EU NIS directive. Available: <http://data.europa.eu/eli/dir/2016/1148/oj>

¹⁴Cybersecurity Act. Available: <http://data.europa.eu/eli/reg/2019/881/oj>

¹⁵Single Digital Gateway regulation. Available: <http://data.europa.eu/eli/reg/2018/1724/oj>

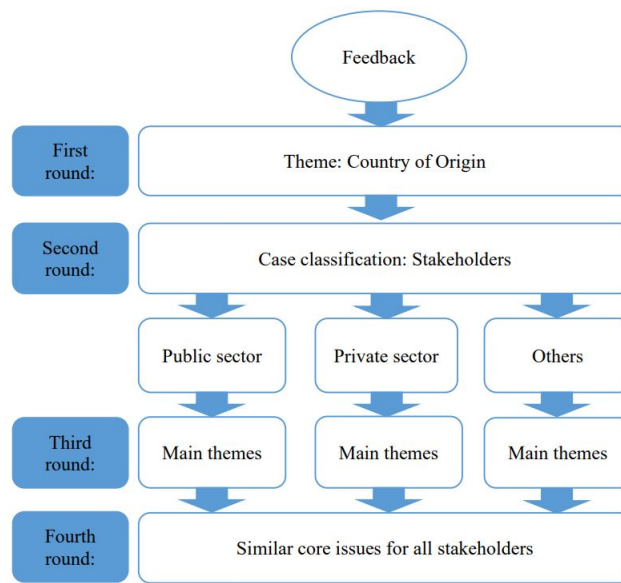


Figure 2: Data analysis model

pillar separately to identify similar problems and core issues for all stakeholders and their possible expectations towards the eIDAS regulation.

5 FINDINGS

During the first round of thematic analysis we identified from which country the feedback was received. Altogether, EC received 53 responses from 16 countries during the public consultation process. Results showed that among the respondents there were representatives of the non-EU countries (Switzerland, UK, USA, Norway), which constituted 19% of all respondents. 7% of the respondents preferred to preserve their anonymity. Therefore, the data about their countries of origin were not available. The largest number of respondents were from France (12), then followed by Germany (7), Belgium (5) and the USA (5). Table 1 represents detailed overview of the respondents by their country of origin.

During the planning phase of the research, we assumed that the largest number of all feedback would be from two types of stakeholders: public and private sector organizations of EU Member States. In contrast, the second round of the data analysis revealed that the third sector organizations, EU citizens and Non-EU organizations actively participated in the consultation process. Therefore, we split the data into three groups by stakeholders: public sector, private sector and others. The following subsections present the research results by each group.

5.1 Private Sector Feedback and Expectations

31 private sector representatives provided their feedback during the public consultation process. 22 of them were EU Member States, 8 responses from non-EU countries and one respondent preferred to stay anonymous.

Business organisations from seven EU countries out of 27 participated directly in the public consultation and sent their feedback.

The largest number of responses came from France (7). German and Belgium organisations sent both five responses each. Concurrently, it is worth of mentioning that business association represented the interests of certain domain companies from a range of countries. Large companies (250 employees and more) constituted 33% of all respondents from EU countries. Small (10 to 49 employees) and micro (1 to 9 employees) organisations contributed equally with a 24% participation rate of all EU companies. Medium companies with 50 to 249 employees amounted to 19% of all respondents from EU countries. All participants from the private sector can be split into two groups: separate companies and various business associations representing interests of different sectors (e.g. Internet and IT services, identification and trust services etc.).

The thematic analysis illustrated that respondents from the private sector emphasised six groups of challenges in the eIDAS implementation process:

- the fragmented legal framework and technical requirements;
- obstacles in mutual recognition and the interaction between the eIDAS-Nodes;
- the limited scope of the eIDAS network;
- security and privacy issues;
- excessive specialisation;
- and a different pace of digitization of the Member States

The most mentioned problems were connected to fragmentation in the legal framework (12 times) and technical requirements (22 times). Since these two themes are intertwined and difficult to split we considered them as one group. In the respondents' opinion, the legal framework needs to be more harmonised on the EU level because the national rules of the Member States stay fragmented and undeveloped. Such fragmentation leads to "a high level of uncertainty for businesses and effectively blocks consumers in some Member States". Besides the fragmented legislation, "the technology,

Table 1: Feedback by country of origin

Country	Number of Respondents	Weighted Percentage
France	12	22,64
Germany	7	13,21
Belgium	5	9,43
USA	5	9,43
Italy	4	7,55
Switzerland	3	5,66
Austria	2	3,77
The Netherlands	2	3,77
Czech Republic	1	1,89
Denmark	1	1,89
Estonia	1	1,89
Finland	1	1,89
Norway	1	1,89
Spain	1	1,89
Sweden	1	1,89
UK	1	1,89
N/A	5	9,43
Total	53	100

eID devices and protocols differ from Member State to Member State". There is also "a lack of common technical standards for digital identity matters". For instance, "the eIDAS does not establish certifiable standards for all digital identity providers". The topic of remote identity proofing and its' lack of harmonization is the most mentioned in this group (11 times).

Approximately the same number of respondents from the private sector see obstacles in mutual recognition and eID schemes notification procedures (16 mentions), with the interaction between the eIDAS-Nodes (5) and in lack of relevant attributes (3). The category related to mutual recognition and eID schemes notification procedures includes the complexity of the notification process, incompatible requirements between policies, different interpretations of some articles of the regulation by national authorities.

Representatives from the non-EU countries would like the EU to collaborate on the international level and enable mutual recognition of the eID schemes. Moreover, two respondents draw attention to the fact that there is a lack of advisory institution on the EU level that is "advisory/administrative body to support the industry by implementing eIDAS". In addition, national "supervisory bodies have no legal enforcing authority". Therefore, "a set of baselines of auditing rules and a baselines audit plan for each trust service" needs to be created. Two respondents brought up the need to cover management of emergencies (including back-up of eID schemes for emergency purposes).

There is also a need to amend the interaction between the eIDAS-Nodes (5 mentions). Identity matching is problematic as "some Member states do not have persistent identifiers", "no access requirements to exchange data between two eIDAS services". Further, the lack of relevant attributes for several services was mentioned three times.

Another group of obstacles is relates to the limited scope of the eIDAS network and lack of demand and use cases, which was

mentioned 21 times. Respondents found that "the current eIDAS framework is restricted to specific use-cases and is not a good fit for many solutions providing digital identity verification, particularly in the private sector". The respondents propose that the framework could be extended to the private sector. Furthermore, more attention should be drawn to user experience and consumer preferences, including authentication processes.

From the security and privacy aspects (16), respondents found that there is a deficit of clarity about the eIDAS levels of assurance and too much interpretation room. Overall, the issue with the level of assurance was mentioned nine times. Some representatives suggested that the eIDAS regulation should be harmonized with the EU Cybersecurity Act and rely on General Data Protection Regulation¹⁶. 14 respondents from the private sector argued that some eIDAS norms are excessively specialized and, in some countries, local regulations are "restrictive and technology-specific". Consequently, the stakeholders warned that excessive regulation might lead to "rapid regulatory obsolescence" and restriction of innovation.

Respondents propose that eIDAS regulation should remain technologically neutral and used solutions must take into account the dynamic and evolving nature of the digital economy and the infrastructure (e.g. endorsing the OpenID Connect Standard besides SAML).

The different pace of digitization across the EU was mentioned three times (e.g. all Member States do not offer eID).

Regarding the EC's options for further eIDAS framework development (1. revise and slightly update the current regulation, 2. extend the effect of eIDAS to the private sector, 3. launch a European Digital Identity (EUID) or combine these three solutions), the preferences of private sector participants were split mainly between various combinations. 36% (11 respondents) of all private

¹⁶General Data Protection Regulation. Available: <http://data.europa.eu/eli/reg/2016/679/oj>

sector respondents, did not choose any option or combination of options. Overall, combinations between options 1, 2 and 3 and their combinations were equally popular.

5.2 Public Sector Feedback and Expectations

Seven representatives from the public sector organizations provided their feedback during the public consultation process (three from the national level, one from local, two from public academic institutions, and one from the postal service provider). Two of them represented French organizations, and others were from Spain, Italy, Estonia, the Netherlands and Finland. It would be better to explore the academic institutions' feedback separately, yet the small number of responses (two) does not allow generalizations. Therefore, public sector stakeholders' expectations also include an opinion of the research institutions.

Overall, the respondents from the public sector found the eIDAS regulation very valuable. However, at the same time, they point out that the legal regulation is not complete and does not cover all important areas, especially from the private sector perspective the whole potential of the regulation is not used.

Public sector representatives see the shift towards the attribute-based approach, but not towards the decentralized architectures, where the storage of attributes is under the direct (physical) control of users. The most frequently mentioned problems were related to the lack of standardization and control (was mentioned in 4 responses). Public sector representatives brought out following aspects regarding standardization:

- it is important to cover transactions between private parties;
- standardize the peer-review procedure;
- specify the minimum criteria relating to remote identification;
- determine the identification of devices and the Internet of Things procedures;
- organize training for citizens.

The revised version of the eIDAS regulation should create a legal grounds for allowing natural and legal persons to use a qualified electronic signature or seal. Some respondents argue that the trust services list should be further expanded or scope broadened (e.g. electronic archiving).

Alternatively, in others' opinion: "The introduction of digital identity trusted services, other than the eIDs already implemented under the eIDAS regulation, should not be pursued. As previously noted, if this were to happen it could undermine the massive efforts, organizational and economic, put in place by the Member States that have already developed notified digital identity systems."

There is also a lack of technological variations of the qualification mechanism or it is too specific, which may lead to the technological neutrality issue that was also brought out by other researchers [27].

When it comes to the three options proposed by the EC (1) revise and slightly update the current regulation, 2) extend the effect of eIDAS to the private sector, 3) launch a European Digital Identity (EUid) or combine these three solutions, respondent opinions were split between the first, second, and combined option. The respondents had different opinions about proposed options on the eIDAS development. Those favouring the first option, were concerned about additional financial costs and organizational changes

of the already existing systems, which the second and third solution might cause. The public sector respondents did not show significant support for the third option due to financial considerations and respondents were afraid of setting up and managing parallel eID systems. They also saw a planned EUid as voluntary option. However, respondents found that the third option may be favourable for legal entities.

Public sector representatives supported the principle that notification of national eID schemes would be mandatory. The respondents were concerned about standardization and privacy related issues. People-centric approach was important for the public sector.

5.3 Third Sector, EU Citizens and Other Stakeholders Feedback and Expectations

The third group of respondents included 15 feedback: 10 from the EU Member States, two from non-EU countries (the USA and Switzerland) and three respondents with unavailable data. The most considerable number of responses in this group were from France (4), followed by respondents from Italy (3), Germany (2) and the Czech Republic (1). Five out of fifteen were NGOs from the identification, trust services and research domains, five EU citizens, one from council of notaries and four respondents preferred to remain anonymous.

The thematic analysis reflected four main challenges in the eIDAS implementation process. This includes the first group with fragmented technical requirements and legal framework, the second group with eIDs mutual recognition and lack of relevant attributes, the third group about the limited scope of use cases and finally security and privacy issues. Challenges from the last two groups were mentioned once in each case.

The most mentioned issue was technical requirements fragmentation (8). Respondents found that more strategic directions are needed for leveraging the benefits for the end-users and the system. Respondents expected the European Standard Organizations to complete the current set of eIDAS. They also emphasised that over regulations should be avoided.

The limited scope of the eIDAS framework was mentioned seven times. As a solution they proposed to stimulate the market and create new trust services and extend the regulation over the private sector entities. Some non-EU participants from this group reminded to consider the cases where EU citizens need to use electronic identities outside of the EU and extend interoperability to the international partners.

Security and privacy topics are essential for this group (5 mentions). For instance, some respondents concerned about private trust service providers, who might not guarantee sufficient personal data security if there are no specific rules and standards to follow.

Lack of relevant attributes was mentioned six times, while obstacles in mutual recognition – three times. Some respondents believed that "the notification process at European level shall remain a prerogative". Respondents proposed harmonisation of legal entity datasets and harmonise the identities of professionals using Legal Entity Identifier (LEI) that enables to link persons, companies and devices.

Almost half of the respondents of this group preferred to notify the EC about their concerns on further eIDAS framework development and not to choose between options proposed by EC.

6 DISCUSSION

Based on the research results it is possible to analyse the EC's European Digital Identity framework proposal and different stakeholders' expectations towards it. Stakeholders see various challenges in the eIDAS implementation and many of them are similar. Among mentioned obstacles are fragmented technical requirements and legal framework, the limited scope of eIDAS and use cases, security and privacy issues, the complexity of the notification procedure and excessive specialization. Those perceived shortcomings correspond with the previous research results that indicated "compliance issues", "interpretation problems", "different practices in member states" and "representation of legal person challenges" [12].

If we analyse the EC's proposal, it is possible to say that the EC has covered many of the stakeholders' expectations. For example, widening the current eIDAS regulation scope in terms of new trust services, harmonizing the regulation with other EU regulations and standards, focusing on the cross-border service provision inside the EU and clarifying sharing of different electronic attributes.

However, there are some important topics that EC proposal does not address brought out by stakeholders. It has to be noted, that the stakeholders emphasized the need of technology neutrality and they did not prefer the option to launch a European Digital Identity (EUId). The EU digital identity wallet proposal seems to be further development of the initially proposed EUId solution, that enables additional benefits for the users (e.g. attribute sharing, qualified digital signature etc.). Stakeholders also expect solution to the legal entities representation issue that the EU digital identity wallet should solve.

It is questionable if the EC proposal should cover standardized e-service provision across the EU borders with third countries. However, the stakeholders are interested about this topic and would like it to be clarified. The EC proposal does not pay much attention to the notification of the national eID schemes mentioned by the stakeholders. However, the notification process does not have to be regulated on the legal act level and can be specified in other regulatory documents.

Security and privacy issues are not directly reflected in the proposal. However, it is possible to take these aspects into account while designing the technical architecture of the EU digital identity wallet and specifying applicable technical standards. Stakeholders also warn against over regulation and standardization that may also happen during the legislative process.

7 RESEARCH LIMITATIONS AND FUTURE DIRECTIONS

Every research has some limitations. For example if it comes to the stakeholders, then the private sector view was more strongly presented than public sector opinion. Further, it is possible to notice that France was very much engaged in the consultation with the most significant share of all participants. The probable explanation of such interest was that France was preparing to pre-notify its eID

scheme under the eIDAS regulation at that time, which resonated through a high participation rate in the consultation process.

On the other hand, the low number of respondents from the public sector and citizens' representatives limit the possibilities to generalize enough the research results of this sector. Moreover, the results reflect only opinions of those, who provided their feedback on the eIDAS public consultation process. We also could only analyse the opinions that were published on the EC website.

Future research perspective should cover analysing the final outcome of the EU Digital Identity Framework regulation. Also, it is possible to evaluate some of the stakeholders expectations after the EC and the Member States have agreed the technical details of the EU digital identity wallet solution.

8 CONCLUSION

The eIDAS revision is a part of the EU strategy, because the EU Digital Single Market largely depends on its enablers: eIDs and electronic trust services. To ensure a high usability of eIDs and to correspond to the users needs, it is important to take into account the feedback of different stakeholders in the eIDAS revision process.

This research focused on the public and private sector stakeholders' feedback analysis provided during the eIDAS public consultation procedure initiated by the EC. We compared the stakeholders feedback with the EC eIDAS amendment proposal and establishing a framework for a European Digital Identity. The aim was to identify if the stakeholders expectations were covered and to contribute to the ongoing eIDAS revision process. Stakeholders mentioned following challenges regarding the eIDAS regulation: fragmented technical requirements and legal framework, the limited scope of eIDAS and use cases, security and privacy issues, the complexity of the notification procedure and excessive specialisation.

Research results indicate that the majority of the stakeholders' expectations are covered by the EC proposal. However, many aspects and their correspondence to the stakeholders' needs depend on the final technical, architectural and procedural agreements between the EC and the Member States. On the stakeholder level, there are some eID related topics that need EU level solutions, but are left out of the EC's proposal. Discussions concerning the EC's proposal are ongoing and final evaluation can be done once the final regulation draft is accepted.

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